

The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of how vaccines have transformed public health by eliminating or drastically reducing deadly diseases over the past two centuries.

Audience: public health professionals and policy analysts

Source: Our World in Data vaccination brief.

100%

US elimination for diphtheria, smallpox, and acute polio cases and deaths

<1 year

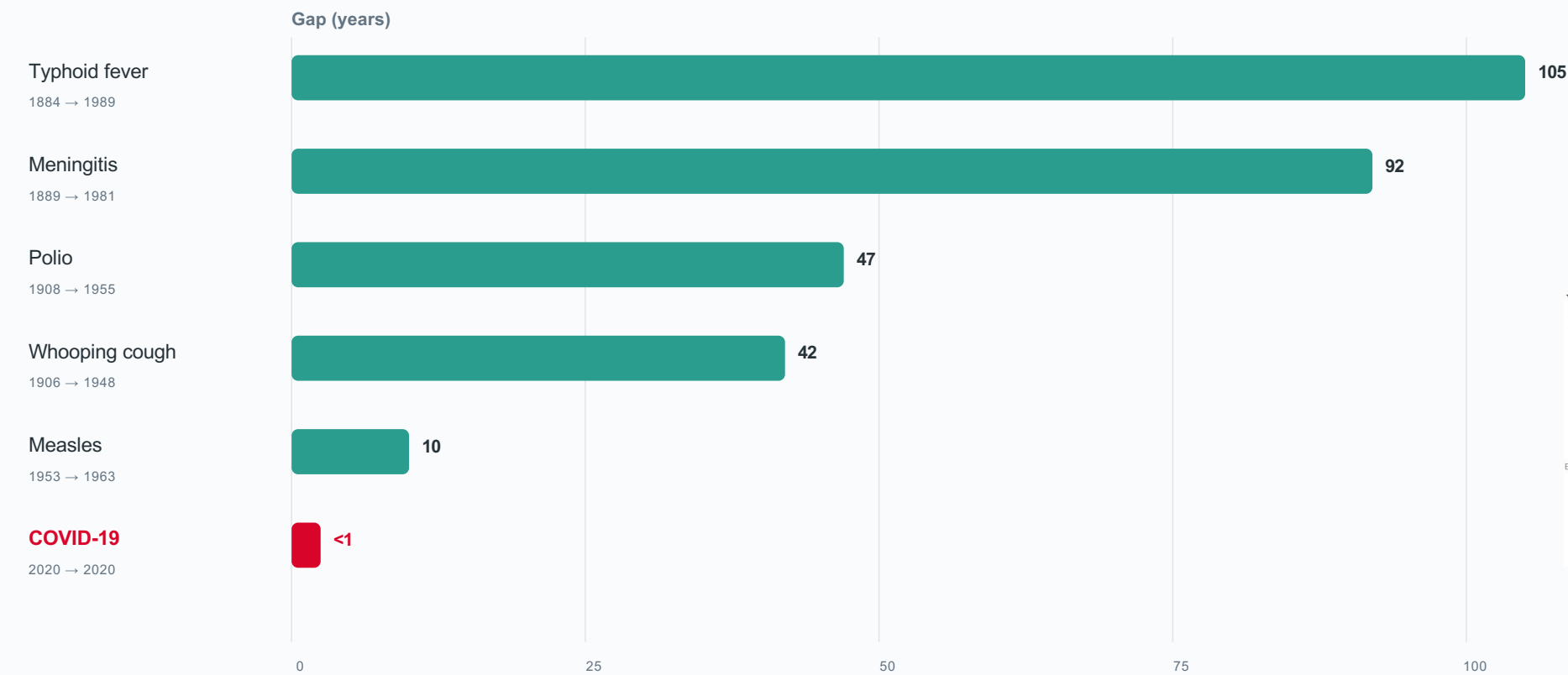
COVID-19: pathogen identified and vaccine licensed in 2020

1980

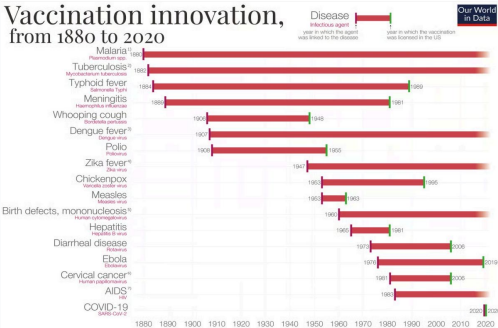
Smallpox eradication certified worldwide

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Time from pathogen identification to vaccine licensure has shortened dramatically over the past 140 years.



Record acceleration
COVID-19 moved from pathogen identification to vaccine licensure in under 1 year.

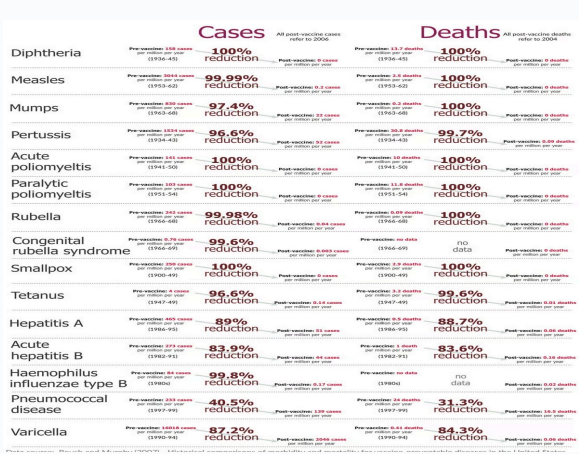


Modern tools cited in the brief: advanced microscopy, genome sequencing, mRNA technology, and virus-like particles.

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Pre-vaccine versus post-vaccine disease burden in annual cases and deaths per million people.

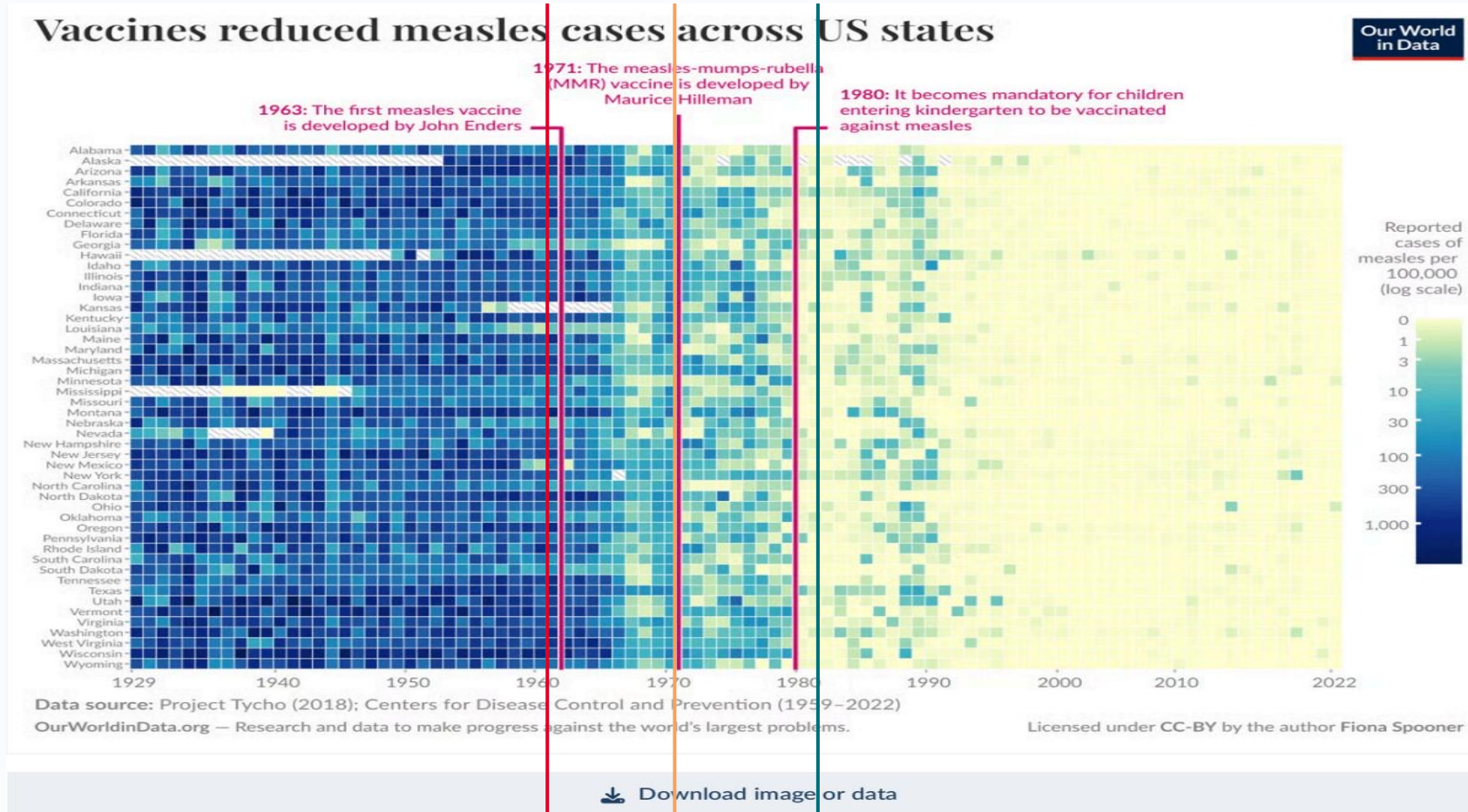
Disease	Pre-vaccine cases	Post-vaccine cases	Case reduction	Pre-vaccine deaths	Death reduction
Diphtheria	158	0	100%	13.7	100%
Measles	3,044	0.2	99.99%	2.5	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%



Remaining burden is concentrated where reductions are incomplete: pneumococcal, hepatitis A, and acute hepatitis B.

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Heatmap-style view of measles cases per 100,000, 1929–2022, with three policy and vaccine milestones.



1963 First measles vaccine developed by John Enders

1971 MMR vaccine developed by Maurice Hilleman

1980 Vaccination became mandatory for children entering kindergarten

Before 1963: high incidence in nearly every state (100–1,000+ cases per 100,000). After 1980: near zero, with only sporadic outbreaks.

Smallpox: The Only Human Disease Eradicated Through Vaccination

A global vaccination campaign moved from WHO resolution to the last natural case and worldwide certification.

1959

1967

1977

1980

WHO passed a resolution to eradicate smallpox

WHO launched an intensified campaign

Last natural case occurred in Somalia

Smallpox declared eradicated worldwide

Why eradication was possible

Virus infected only humans

Symptoms were distinguishable

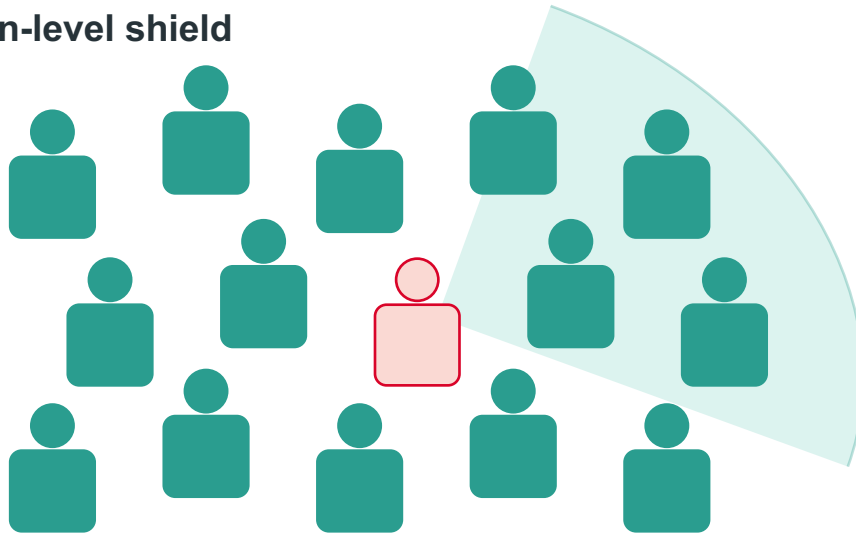
Vaccine was cheap and effective

Smallpox remains the only human disease to be fully eliminated through vaccination.

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

When enough people are vaccinated, pathogens cannot spread effectively through the population.

Population-level shield



Vulnerable individual shielded by surrounding vaccination

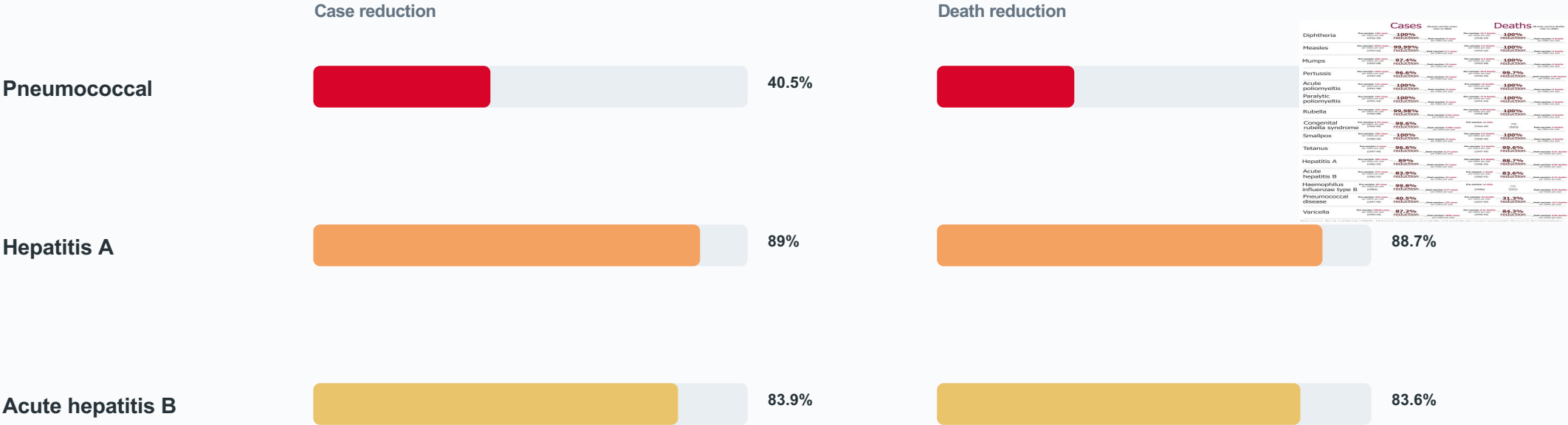
Vaccinated

Vulnerable

- Vaccines protect individuals and entire communities.
- When enough people are vaccinated, pathogens cannot spread effectively.
- This shields those who cannot be vaccinated: newborns, immunocompromised individuals, and cancer patients on chemotherapy.

Gaps Persist: Pneumococcal Disease Shows Only 40.5% Case Reduction

Vaccination progress is dramatic, but incomplete control remains for pneumococcal disease and hepatitis.



Barriers named in the brief: not everyone has access to vaccines globally, supply shortages put children at risk, and poor scientific understanding can reduce trust and uptake.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

100%

Elimination of cases and deaths for diphtheria, smallpox, and both forms of polio in the US

>100 → <1

Development timelines collapsed from over 100 years for typhoid to under 1 year for COVID-19

1980

Measles cases dropped to near zero after the kindergarten vaccination mandate

Only one

Smallpox remains the only human disease eradicated through vaccination

Gaps persist

Pneumococcal disease, hepatitis A, and hepatitis B show incomplete reductions

Call to action for global health systems: improve access, coverage, and trust in vaccination.